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Hadamard Gate in Qiskit
Video: Hadamard Gate in Qiskit
3 min
Graded Assignment: Assessment 6
Submitted

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Assessment 6

Review Learning Objectives

Coach
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Assignment details
Due Dec 18, 11:59 PM IST
Attempts Unlimited
Submitted Dec 10, 2:20 AM IST

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To pass you need at least 80%. We keep your highest score.
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1. Select all the statements that correctly describe properties of eigenvectors and eigenvalues in the context of quantum computing. **1 / 1 point**

- [x] In quantum computing, eigenvalues are used to find measurable quantities of observables.

 Correct
 Correct! Eigenvalues help in determining observable quantities in quantum systems.

- [x] Eigenvalues can provide insights into the stability of quantum systems.

 Correct
 Correct! Eigenvalues are important in understanding system stability.

- [] If a matrix has an eigenvalue of zero, then its eigenvectors form the null space of the matrix.
- [] Eigenvectors are always orthogonal to each other.
- [x] Eigenvectors remain invariant under their associated linear transformation, except for scaling by the eigenvalue.

 Correct
 Correct! This is a fundamental property of eigenvectors.

2. Which of the following best describes the significance of eigenvectors in transformation matrices? **1 / 1 point**

- (•) They indicate the directions along which linear transformations act by stretching or compressing.
- () They determine the speed at which a quantum computer performs calculations.
- () They are used to encode data into qubits for quantum computations.
- () They specify the probability distributions of quantum states.

Correct
Correct! Eigenvectors show the directions in which a transformation has a straightforward effect.

3. Which gate is used to create superposition in a qubit? **1 / 1 point**

- (•) Hadamard gate
- () Pauli-Y gate
- () Pauli-X gate
- () CNOT gate

Correct
Correct! The Hadamard gate is used for this purpose.

4. What is the primary effect of an X gate on a qubit initially in state $|0\rangle$?

- (•) It flips the qubit's state to $|1\rangle$.
- () It creates a superposition of $|0\rangle$ and $|1\rangle$.
- () It leaves the qubit in state $|0\rangle$.
- () It entangles the qubit with another qubit.

Correct
Correct! The X gate, also known as a NOT gate, flips the qubit's state from $|0\rangle$ to $|1\rangle$.

5. Which quantum gate can be used to create an equal superposition of $|0\rangle$ and $|1\rangle$? **1 / 1 point**

- (•) Hadamard gate
- () Z gate
- () Y gate
- () X gate

Correct
Correct! The Hadamard gate creates an equal superposition of $|0\rangle$ and $|1\rangle$ states.